

AEM 8232 – Spring 2026
Molecular Gas Dynamics II

HOMEWORK #1

Due: Thursday, February 5th

PROBLEM 1.

(a) Write a computer program that generates a file of particle velocity vectors sampled from a Maxwellian velocity distribution for an Argon gas. Note these are the actual particle velocities (*not* just the thermal velocities).

A typical file format may look like:

```
-----  
Cx   Cy   Cz   - basically just one line for each particle  
Cx   Cy   Cz  
...  
...  
Cx   Cy   Cz  
-----
```

Thus, given a total number of desired particles (N), a bulk velocity vector $\vec{V} = (u, v, w)$, and a temperature T , your program should generate a large file of individual particle velocities sampled from the corresponding Maxwellian distribution.

(b) Next write a separate program that reads in the file created in (a) and then:

- (i) Computes the bulk velocity vector, $\vec{V} = (u, v, w)$, by averaging over all particle velocities in the file.
- (ii) Computes the temperature, T , by the averaging appropriately over all particle velocities in the file.
- (iii) Computes the velocity-distribution-function (vdf) by binning particle velocities into small bins, counting the number of particles in each bin, and plots the resulting 'histogram' (i.e. plotting the vdf).

(c) Present the results for computed $\vec{V} = (u, v, w)$, and T , as well as the normalized vdf (you need only plot the C_x component) corresponding to the following $(\vec{V}, T)$ conditions:

- high $\vec{V}$ - high T ($u=1$ km/s, 1000K)
- high $\vec{V}$ - low T ($u=1$ km/s, 100K)
- low $\vec{V}$ - high T ($u=100$ m/s, 1000K)
- low $\vec{V}$ - low T ($u=100$ m/s, 100K)

[You can take $v = w = 0$ for these examples, but your code should be general.]

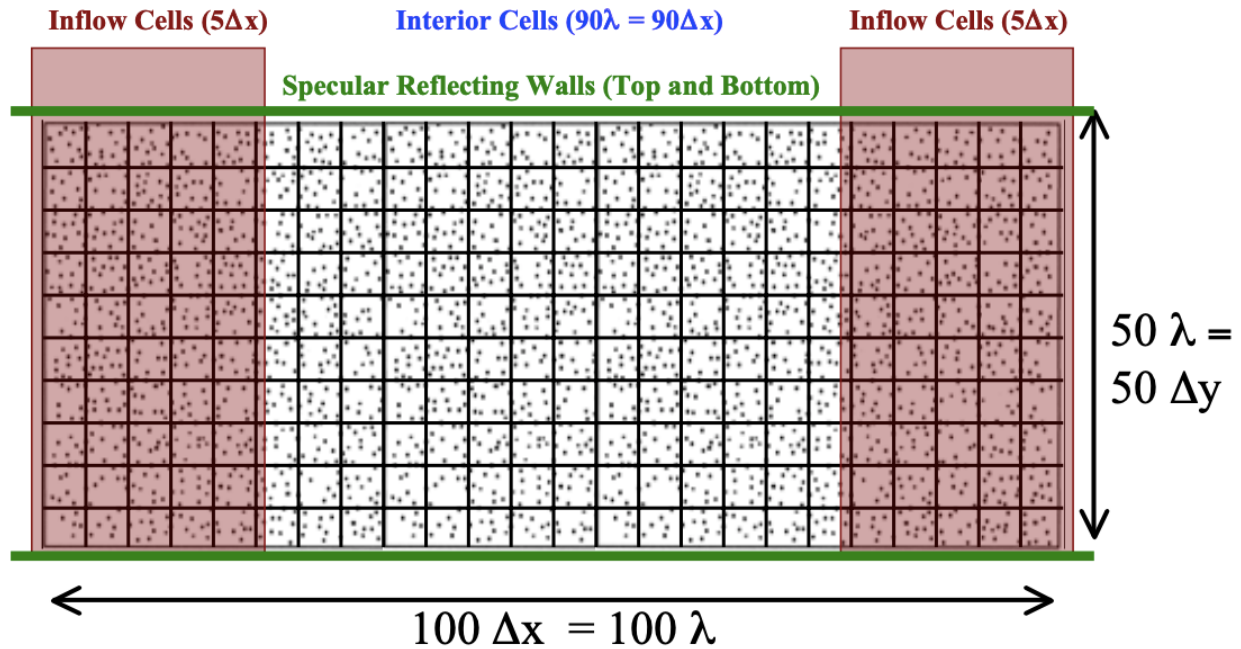
Finally, generate the results for (c) using few particles ($N=100$) and many particles ($N=500,000$). When plotting the computed vdfs, always overlay them with the exact analytical Maxwellian curve that you are trying to generate. Discuss the results.

PROBLEM 2.

Incorporate your particle generation and sampling codes into the MGDS-Lite code and simulate a *free-molecular*, uniform, flow in 2D.

HAND IN YOUR CODE (likely subroutines that plug-in to MGDS-Lite).
PLEASE KEEP CODE LISTINGS COMPACT (SMALL FONT, SMALL SPACING, ETC.).

The computational domain and simulation setup look like the following:



Flow Conditions (~80km altitude):

Nitrogen gas (N_2), $\rho = 1.6 \times 10^{-5} \text{ kg/m}^3$, $T = 200 \text{ Kelvin}$, $u = 50 \text{ m/s}$ ($v=w=0 \text{ m/s}$)

Notes:

Particles should be given C_x , C_y , and C_z velocities. However, if MGDS-Lite 2D-mode is used, then although C_z will be assigned to each particle, its position in z is irrelevant. Particles are sorted into cells based on their x and y position only.

Assume a Hard-Sphere model with hard-sphere diameter of $d = 4 \times 10^{-10} \text{ m}$, in order to approximate the mean-free-path (λ) and the mean-collision-time (τ_c):

Eq. 1.144 in textbook $\rightarrow \lambda = 1/(n\sqrt{2}\pi d^2)$

Set the simulation particle weight (W_p) so that you have ~20 particles per cell. This means that you will have approximately 100,000 particles once the entire domain is filled, based on the above figure. The cell sizes should be set as $\Delta x = \Delta y = \lambda$ and the simulation timestep should be constant and set to $\Delta t = \tau_c / 5$.

Results:

- a) Plot the total number of particles (N_{total}) vs. the number of timesteps. You should see N_{total} increase from zero and reach a steady-state value close to 100,000. Use your plot to determine how many timesteps are required to reach steady-state. Could you have predicted this value? Explain briefly.
- b) After steady-state has been reached, determine the value of u and T in one cell (somewhere near the middle of the domain) and plot u and T at each Δt for 100 timesteps. That is plot u vs. time and T vs. time calculated within a single cell. How much scatter do you see in the results (% of the mean value)?
- c) After steady-state has been reached, determine the value of u and T in one cell (somewhere near the middle of the domain) using particles *accumulated* over 100 timesteps. That is, determine a single steady-state value for u and T within that cell. Comment on your result compared to the result in (2).
- d) After steady-state has been reached, determine and plot the vdf in one cell (somewhere near the middle of the domain) using particles *accumulated* over 10,000 timesteps. You may bin particle velocities during the calculation. Or you may simply dump all particle velocities in the cell to a file, which is appended at each timestep, and process the file after the simulation is complete. Comment on your vdf plot and compare with the expected analytical distribution.
- e) After steady-state has been reached, determine the value of u and T in *all* cells using particles *accumulated* over 100 timesteps. Create contour plots of the velocity and temperature field in the 2D domain. Repeat this twice more: Once by sampling over only 10 timesteps, and another by sampling over 1000 (or even 10,000 timesteps – depending on simulation time). Discuss your results briefly.
- f) Repeat (e) but this time *do not* generate particles in the *downstream* Inflow Cell region. Simply delete particles that leave the downstream boundary, but do not generate particles in that region. Sample your solution for at least 100 timesteps. Plot contours of u and T for the 2D domain. Explain your results.